Entropy production in nonreciprocal networks of decentralized LLM agents with coupled beliefs and actions

Anonymous working draft

V10 formulation study, August 17, 2026

Abstract

We study a two-layer stochastic model in which persistent decentralized agents maintain distinct binary beliefs and committed actions on communication and dependency networks. The reciprocal model is a finite reversible heat-bath chain. For a directed perturbation $W_\alpha = W_0 + \alpha V + O(\alpha^2)$, we derive the first stationary-response correction and show that stationary entropy production has the onset $\sigma(\alpha) = C\alpha^2 + O(\alpha^3)$, with C an explicit non-negative quadratic form in the derivative of equilibrium probability flux. Exact enumeration and larger pathwise simulations test the coefficient, its normalization, and its dependence on topology, temperature, system size, and belief–action coupling. We also test an executable realization in which independent Qwen agents own private observations, memories, inboxes, beliefs, actions, and typed authority; the scheduler supplies update opportunities but does not choose their states. Across 384 retained pilot decisions, structured validity and response to private evidence were adequate, but agents did not change belief in any of 48 final matched counterfactual pairs that differed only in a newly delivered peer message. This failed the frozen message-use prerequisite, so no formal empirical kernel or dynamic LLM-network study was run. The entropy-production conclusions therefore concern the analytical stochastic-agent model; the LLM pilot supplies a boundary result, not an LLM entropy-production measurement.

1 Introduction

Local stochastic decision rules connect statistical mechanics with strategic interaction and decentralized coordination [1, 2]. Directed communication creates a more delicate problem: asymmetric influence generally destroys a potential representation and supports stationary probability currents. Their dissipation is naturally quantified with Markov network entropy production [3, 4]. Nonreciprocal collective systems and asymmetric kinetic Ising models already show rich nonequilibrium behavior [5, 6]; recent work also establishes a quadratic near-reciprocal entropy-production response in a metric kinetic Ising model [7]. The quadratic onset itself is therefore not a novelty claim here.

Our narrower theoretical contribution is an explicit stationary-response coefficient for a coupled belief–action finite Markov chain, including a decomposition by the variable layer changed along a transition. Our empirical objective is to determine whether the same construction can be measured in actual decentralized language-model agents. Recent work already places LLM populations in binary statistical-physics experiments [8]; the substantive distinction sought here is persistent private state, separate belief and executed action, directed natural-language messages, and qualified time-reversal inference. The pilot reported below shows why that objective has not yet been met: under this interface, delivered peer evidence did not alter the controlled belief state, and the formal experiment stopped prospectively.

2 Relation to prior statistical-mechanical models

The two-variable construction is mathematically close to an Ising bilayer with an on-site interlayer coupling; equilibrium phase diagrams for bilayer Ising systems long predate this work [9]. Its interpretation differs: the layers represent a private belief and a consequential committed action, and only the belief layer receives directed natural-language influence. Multiplex network notation formalizes the distinct edge sets [10], but does not by itself supply a nonequilibrium thermodynamics.

Persistent memory is also not new to kinetic Ising dynamics. Self-interaction can change detailed-balance properties depending on sequential versus parallel updating [11]. V10 therefore treats LLM memory as a possible hidden state to diagnose, not as an innocuous embellishment. Likewise, asymmetric kinetic Ising mean-field theory already relates directed couplings to delayed correlations and entropy production [6], while perturbative entropy production has been derived for other nonreciprocal particle systems [12]. The finite-kernel response in equations (9)–(13) is valuable for its exact indexing, normalization, and layer decomposition, not because perturbation around reciprocity is unprecedented.

In strategic settings, logit updates recover Gibbs laws only when incentives derive from an appropriate potential [2]. Directed influence is a controlled way to leave that potential class. For partially observed paths, time-reversal divergence supplies a lower bound rather than the total hidden entropy production [13, 14]. These precedents determine the terminology and null controls used for the LLM stage.

3 Decentralized agent architecture

Agent i owns a private observation stream, memory M_i, belief b_i, action a_i, commitment state, inbox, outbox, role, and typed authority. A message is available only after delivery over a directed edge. The environment schedules an update opportunity and validates a returned structured action; it may not replace the agent's choice. Counterfactual isolation tests change one vault without mutating another. Figure 1 distinguishes this deployable boundary from evaluator-only state reconstruction.

In controlled micro-update mode, one persistent agent is invited to reconsider either b_i or a_i while retaining the other variable. In autonomous-turn mode it can revise both, send a directed message, maintain or reject a commitment, request information, defer, or use a role-authorized tool. Structured output records the belief choice and confidence, action choice, commitment, signal, message, tool action, and reason code. Natural-language transcripts are external raw artifacts rather than state labels inferred by free-text parsing.

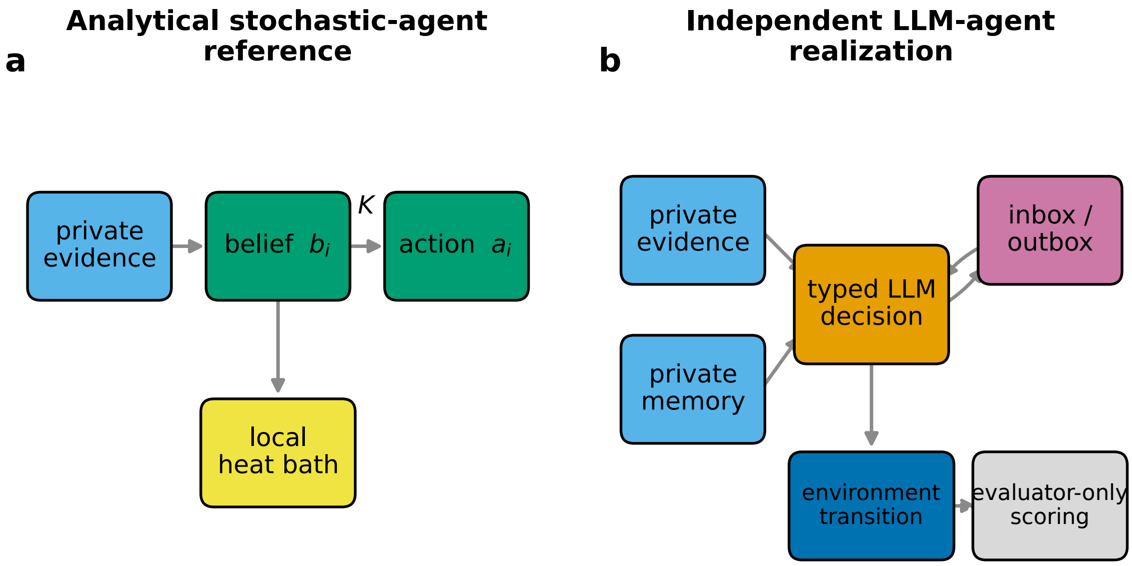

Figure 1: Theory-to-LLM architecture. Filled pathways denote authorized local information and explicit message delivery. The evaluator reconstructs global currents offline but does not select an agent's belief or action.

4 Coupled belief–action model

The analytical microstate is

$$x = (\boldsymbol{b}, \boldsymbol{a}), \qquad b_i, a_i \in \{-1, +1\}. \tag{1}$$

For symmetric communication and dependency matrices $A^{(c)}$ and $A^{(d)}$, we use the dimensionless Hamiltonian

$$\begin{aligned} H(x) = & -\frac{J_b}{2}\sum_{ij} A_{ij}^{(c)} b_i b_j - \frac{J_a}{2}\sum_{ij} A_{ij}^{(d)} a_i a_j - K\sum_i a_i b_i \\ & -\sum_i h_i b_i - \sum_i g_i a_i. \end{aligned} \tag{2}$$

Here h_i is private evidence and g_i a local task drive. These are statistical-mechanical energies only in this explicitly reversible reference; outside it, equation (2) is a symmetric-reference diagnostic.

One of the $2N$ variables is selected uniformly at each attempted update. Its heat-bath probability is

$$\Pr(b_i' = s \mid \mathcal{F}_i) = \frac{\exp[s\ell_i^{(b)}/T]}{2\cosh(\ell_i^{(b)}/T)}, \qquad \ell_i^{(b)} = J_b \sum_j A_{ij}^{(c)} b_j + K a_i + h_i, \tag{3}$$

$$\Pr(a_i' = s \mid \mathcal{F}_i) = \frac{\exp[s\ell_i^{(a)}/T]}{2\cosh(\ell_i^{(a)}/T)}, \qquad \ell_i^{(a)} = J_a \sum_j A_{ij}^{(d)} a_j + K b_i + g_i. \tag{4}$$

The analytical decision temperature T is policy stochasticity. It is not the LLM token-sampling temperature; an LLM effective decision temperature must be fitted from controlled responses.

5 Reciprocal equilibrium reference

If both layers are static and symmetric and all agents share T, states x, y differing in one selected variable obey

$$\frac{W_0(x,y)}{W_0(y,x)} = \exp\left[-\frac{H(y)-H(x)}{T}\right]. \tag{5}$$

Thus

$$\pi_0(x) = Z^{-1}\exp[-H(x)/T] \tag{6}$$

satisfies detailed balance. Self transitions are retained, and one sweep is $2N$ attempted updates. Exact enumeration compares this law with the stationary left eigenvector and verifies the reciprocal entropy-production null. The V10 maximum reciprocal rate is 1.10×10^{-31} nats per attempted update.

6 Nonreciprocity and perturbative entropy production

We orient every edge pair without changing its support or pairwise total weight:

$$A^{(c)}(\alpha) = A_s + \alpha A_a, \qquad A_s^\top = A_s, \quad A_a^\top = -A_a, \tag{7}$$

with weights proportional to $1 \pm \alpha$. This preserves directed message opportunities, edge count, and global total weight, although a particular orientation need not preserve each node's in/out strength. Diagnostics report that residual rather than silently treating it as controlled.

Let

$$W_\alpha = W_0 + \alpha V + O(\alpha^2), \qquad \pi_\alpha = \pi_0 + \alpha r + O(\alpha^2). \tag{8}$$

Differentiating $\pi_\alpha W_\alpha = \pi_\alpha$ and normalization yields

$$r(I - W_0) = \pi_0 V, \qquad r\mathbf{1} = 0. \tag{9}$$

The constrained system fixes the first stationary correction (equivalently it can be written using a group inverse [15]). Define the equilibrium flux $q_{xy} = \pi_{0x}W_{0xy}$ and its first derivative

$$f_{xy} = r_x W_{0xy} + \pi_{0x}V_{xy}, \qquad j_{xy} = f_{xy} - f_{yx}. \tag{10}$$

At detailed balance, both the current and its conjugate log affinity vanish. Their leading terms are first order, so the physical linear term is absent and the Schnakenberg expression

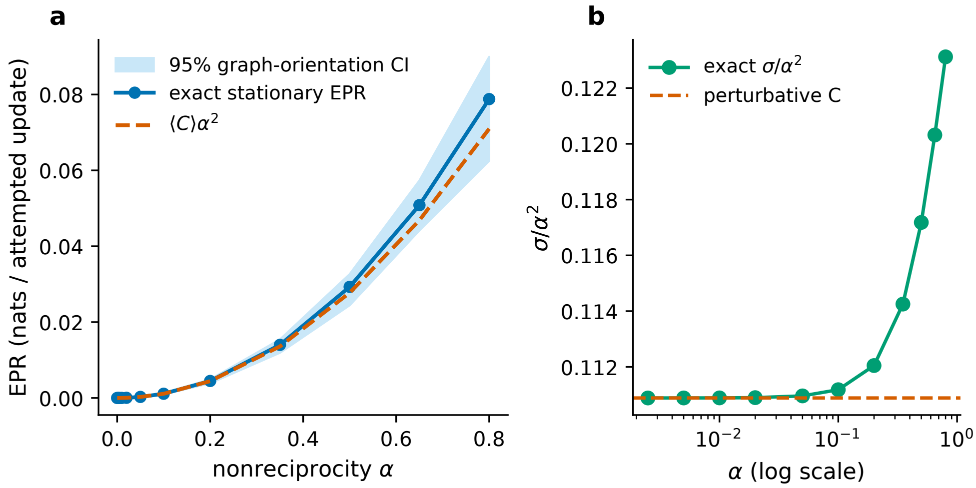

Figure 2: Exact stationary entropy production and the perturbative quadratic prediction. Intervals are across independent directed orientations. The residual panel tests the expansion rather than selecting a curve visually.

expands as

$$\sigma(\alpha) = \frac{1}{2}\sum_{xy} J_{xy}(\alpha)\log\frac{\pi_{\alpha x}W_{\alpha,xy}}{\pi_{\alpha y}W_{\alpha,yx}} \tag{11}$$

$$= C\alpha^2 + O(\alpha^3), \tag{12}$$

$$C = \frac{1}{2}\sum_{x,y:q_{xy}>0}\frac{j_{xy}^2}{q_{xy}} \geq 0. \tag{13}$$

This is an exact perturbative result for the specified finite discrete-time kernel. It is not a claim about arbitrary hidden-memory LLM dynamics.

7 Exact and trajectory methods

For small N we enumerate all 2^{2N} states, construct the row-stochastic kernel, solve for its stationary law, and evaluate equation (13). Independent central differences validate V and finite differences validate C. Figure 2 compares exact rates and predictions across eight independent edge orientations. The maximum relative coefficient error for $0 < \alpha \leq 0.02$ is 5.63×10^{-4} (0.0563%).

The parameter grid varies reciprocal graph topology, T, and K. The predicted coefficient spans 0.00364–0.16628 nats per attempted update. Figure 3 shows its temperature and coupling dependence. The exploratory correlation with the antisymmetric spectral norm is -0.221; it is not a causal topological law.

For larger networks, random sequential local updates produce stationary paths. We average the local log forward/reverse rate ratio and report it in three explicit units: per attempted variable update, per agent-sweep (twice the per-update value), and total per sweep ($2N$ times the per-update value). Independent graph/trajectory seeds, not correlated time steps, are inferential units. A known three-state cycle establishes estimator bias and

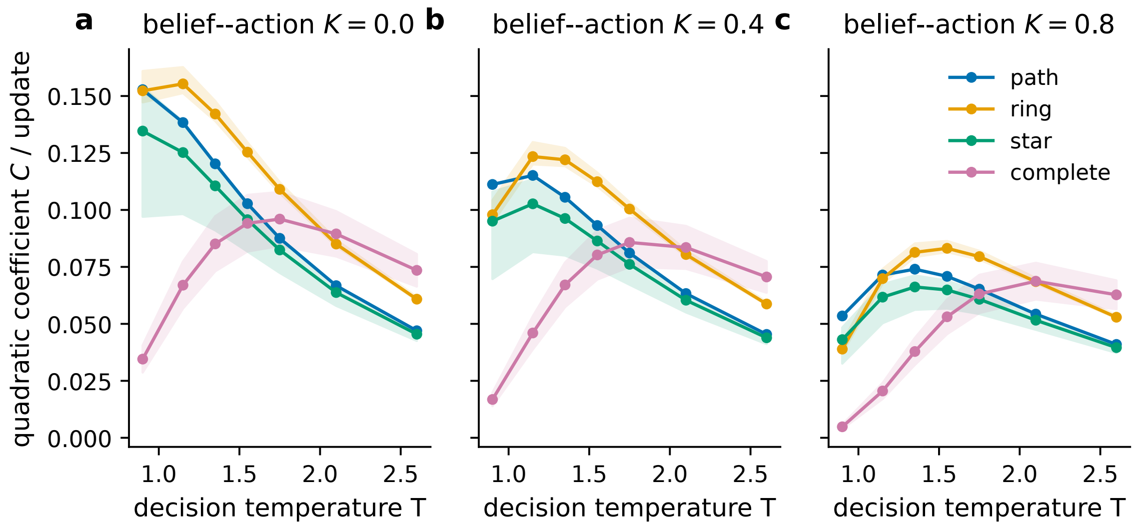

Figure 3: Perturbative coefficient across topology, decision temperature, and belief–action coupling. Points summarize independent orientations.

convergence.

Figure 5 visualizes probability currents in state space. The reciprocal condition has no resolved circulation; directed communication creates cycles spanning both belief- and action-flip transitions.

8 Empirical LLM protocol and identifiability

The primary model is Qwen2.5–7B–Instruct at a pinned revision [16]. Calibration counterbalances plan labels and prompt paraphrases, varies natural-language evidence balance, and fits symmetric and asymmetric logistic responses. Effective decision temperature is inferred only if held-out calibration supports a heat-bath approximation. The planned controlled kernel uses repeated independently seeded decisions from complete small microstates. Larger persistent trajectories use reciprocal and paired nonreciprocal graphs.

Observed $(\boldsymbol{b}, \boldsymbol{a})$ need not be Markov because memory, recent messages, and commitment age may retain predictive information. Conditional mutual information and likelihood comparisons test that adequacy. If it fails, we augment the state where feasible; otherwise block time-reversal KL is reported only as a coarse-grained irreversibility lower bound [13, 14]. Reciprocal, time-shuffled, message-shuffled, communication-disabled, and randomized forward/reverse controls define an empirical bias floor. A positive plug-in number alone is not evidence of full thermodynamic entropy production.

This stage is currently pilot no-go; formal network study not unlocked. The direct public-IP SSH alias was stale, but the established RunPod proxy reached the existing RTX 4090 Pod. Five retained pilot attempts produced 384 structured decisions and 388 model calls including repairs. The corrected private-evidence pilot had a positive-minus-negative response difference of 0.35 and an option-order coefficient of -0.116, inside the predeclared ± 0.15 bound. That result used the prompt immediately before its one message-interpretation clarification and is not a final held-out effective-temperature estimate.

The decisive matched message pilot balanced application, paraphrase, option order,

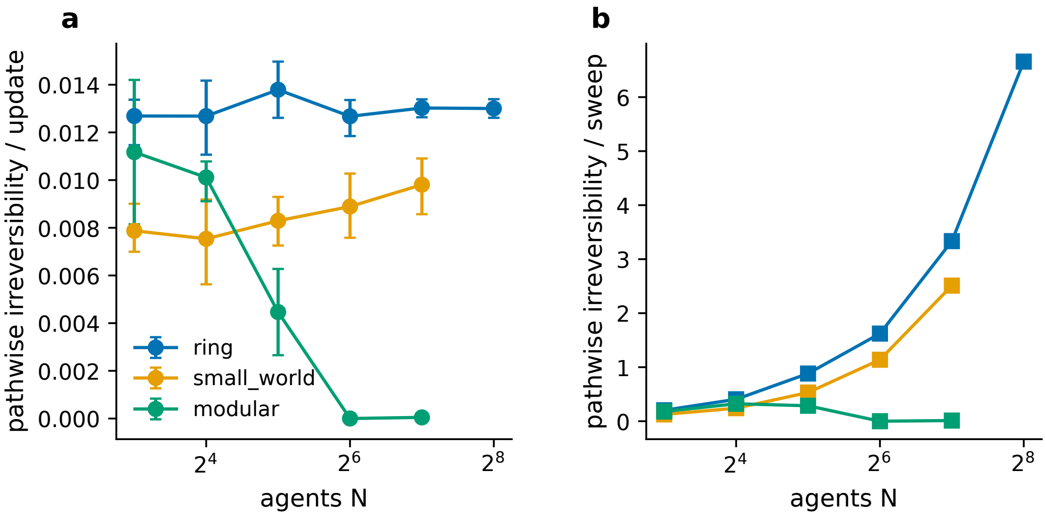

Figure 4: Pathwise stationary irreversibility versus system size. Total, per-agent-sweep, and per-update normalizations prevent an apparent scaling claim from being an artifact of the update convention.

previous belief, and previous action. Before the clarification, every decision retained its prior. After explicitly defining inbox entries as new evidence and the locally visible influence weight as reliability, all 48 final pairs again retained their prior. The right-support minus left-support response was 0.00, below the frozen 0.20 gate. Figure 7 shows both the private-evidence response and this boundary result. Because the intended nonreciprocity manipulation acts through delivered evidence and influence weights, proceeding to a large network study would not test the proposed mechanism. Empirical transition currents, Markov adequacy, and cross-size replication therefore remain untested.

9 Illustrative application mappings

Humanitarian coordination maps b_i to a local belief about two competing allocation/routing plans and a_i to the plan actually committed by a depot, carrier, or field coordinator. Defensive utility restoration maps the same variables to local restoration or isolation plans held by component operators, crew coordinators, relays, and safety monitors. Asymmetric trust, one-way loss, or delayed acknowledgment induces directed communication without exposing global truth. Figure 8 records consequential typed plan effects, but these mappings are illustrative and provide no field-validity or human-benefit evidence.

10 Discussion and limitations

The coefficient in equation (13) explains the quadratic onset without post-hoc curve selection and shows how a communication perturbation can produce currents in executed actions. It also clarifies what is not established. The finite grids do not prove a thermodynamic-limit phase transition. Pairwise weight preservation does not guarantee nodewise degree-strength matching. The pathwise estimator assumes stationary fully specified local rates. Most importantly, LLM inference stopped at qualification: it did not produce a formal transition

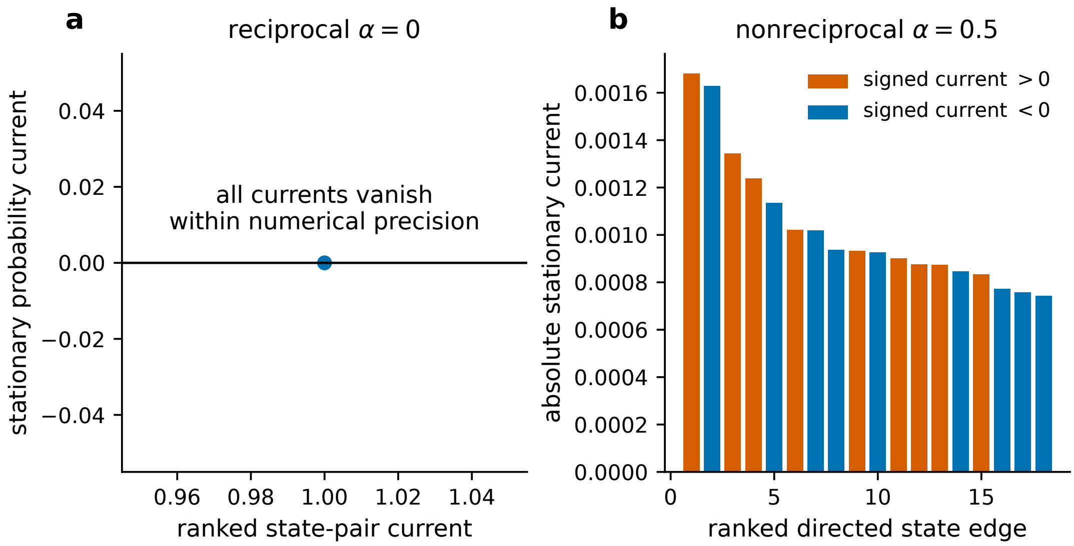

Figure 5: Largest stationary state-space currents under reciprocal and nonreciprocal communication. Action-layer currents can arise from a communication-layer perturbation through K.

kernel or persistent reciprocal/nonreciprocal trajectories. The title therefore remains a prospective research target rather than a completed empirical claim. The present package is a rigorous theory study with an informative LLM interface no-go, not yet the proposed JSTAT LLM-agent paper.

The novelty audit further rules out broad priority claims: quadratic entropy production near reciprocity, asymmetric kinetic Ising dynamics, and binary LLM population experiments all have direct precedents. A defensible eventual contribution requires the combination of coupled belief/action states, private agent memory and messages, typed consequential action, controlled empirical transition kernels, and bias-qualified irreversibility.

11 Conclusion

Around a reversible coupled belief–action reference, nonreciprocal influence has an explicit non-negative quadratic entropy-production coefficient. Exact finite-state computations and pathwise simulations test that statement with unambiguous update units. Whether independently acting language-model agents realize the same response remains open. This V10 implementation did not pass the prior requirement that delivered peer evidence alter local belief, and correctly stopped before estimating network entropy production.

Reproduction and model-compute statement

Compact source code, configurations, aggregate tables, figure source data, and vector figures are repository-facing. Exact raw formal artifacts and checksums are external. CPU reproduction runs audit, tests, development, freeze, formal simulation, analysis, PDF QA, manuscript build, and clean export in that order. The Qwen scripts are opt-in and

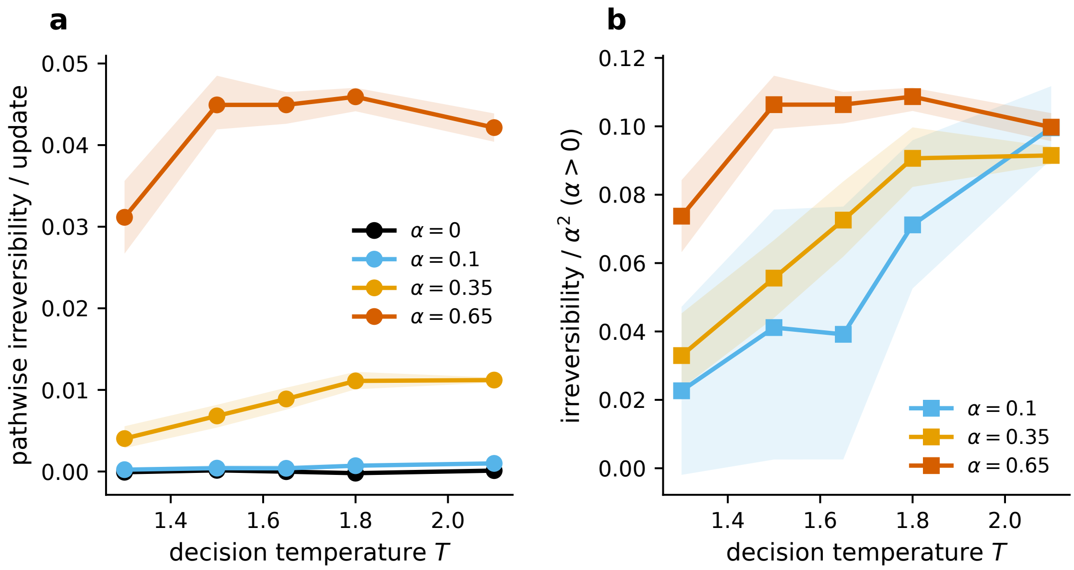

Figure 6: Temperature–nonreciprocity response for larger stochastic-agent trajectories. These finite-size surfaces are not presented as phase diagrams or thermodynamic-limit transitions.

require the pinned model cache on the existing authorized GPU host; they cannot silently fall back to fabricated or mock model evidence. Raw transcripts remain external with checksums. The formal script is locked by both a message-response gate and a source-freeze manifest, neither of which passed/exists in this no-go package. No real human participants or operational infrastructure were used.

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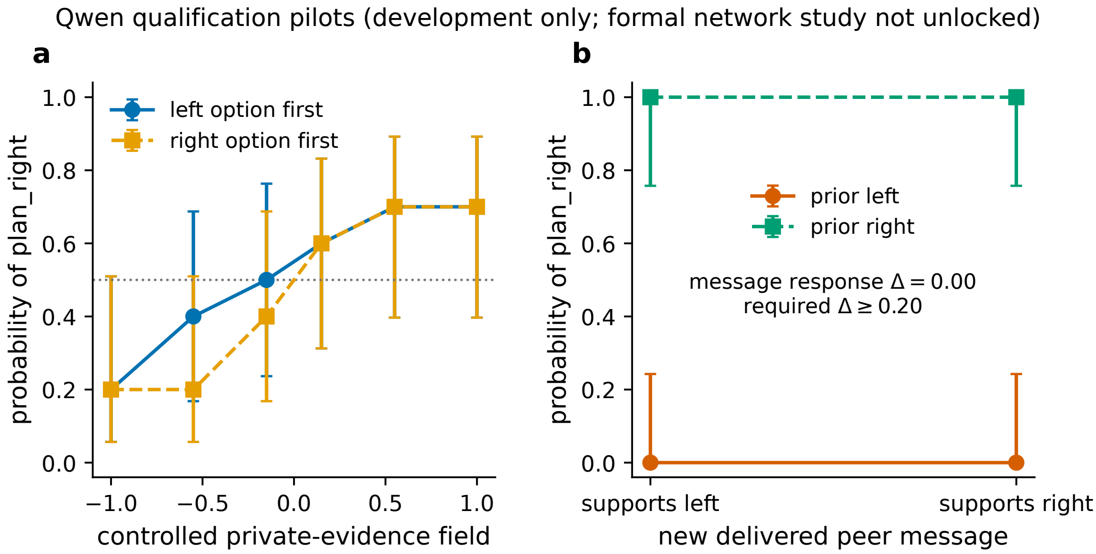

Figure 7: Development-only Qwen qualification. (a) The fully crossed private-evidence pilot produced a graded response with bounded option-order effect. (b) Under the final prompt, changing only the direction of a newly delivered peer message never changed belief; Wilson intervals describe the finite matched-pilot cells. The formal network study was not unlocked.

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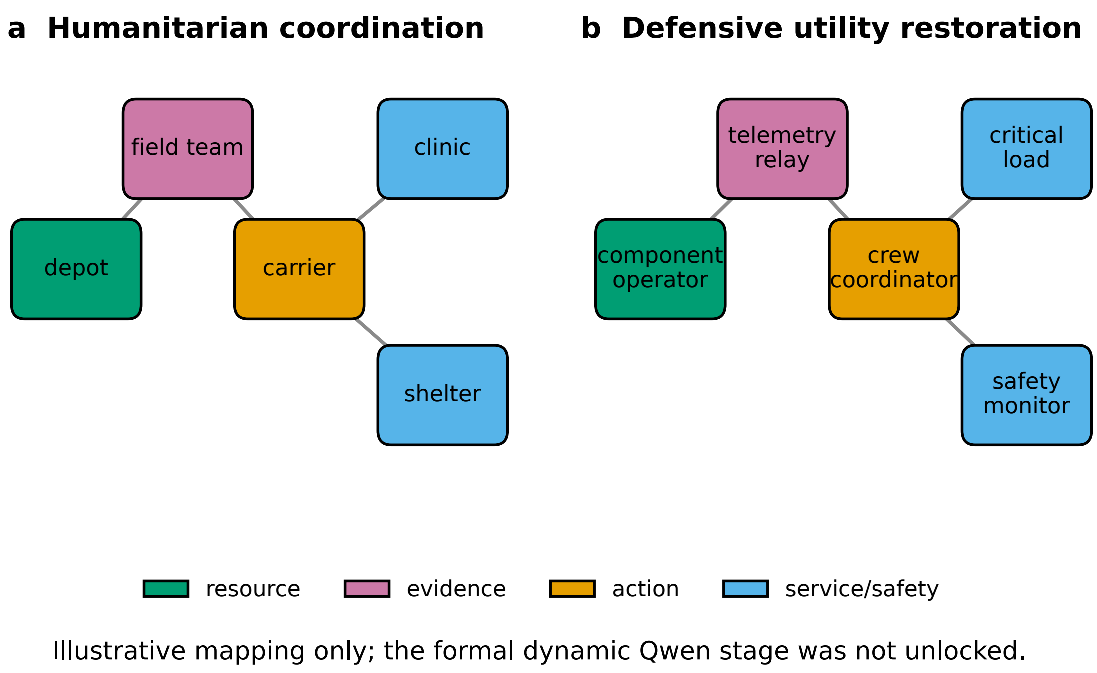

Figure 8: Secondary humanitarian and defensive utility mappings. The same abstract microstate is given domain-specific typed consequences; no offensive cyber procedure or real infrastructure target is represented.

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